

Sagittal growth of the facial skeleton of 6-year-old children with a complete unilateral cleft of lip, alveolus and palate treated with two different protocols

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SUMMARY. Objective: The purpose of the study was to compare sagittal growth of the facial skeleton of 6-year-old children treated in two cleft centres with different surgical protocols. Material and methods: Each group consisted of 20 consecutive non-syndromic children with complete unilateral cleft lip, alveolus and palate. They all had presurgical orthopaedics with a passive plate and external strapping until lip repair. Centre 1 had lip repair at the age of 3 months and one stage palatal closure at the age of 1 year. Closure of the alveolar cleft was planned at 9 years with bone grafting. In centre 2 lip repair was performed at the age of 6 months, soft palate repair at 12 months and hard palate repair together with mucoperiosteal closure of the alveolar cleft at the age of 30 months. At the time of investigation, the children from both centres had not received any postoperative orthodontic treatment. Sagittal growth was evaluated on lateral cephalograms using the angles SNA, SNB, ANB and SNPg. For control, Droschl standards were used. The Mann–Whitney *U* test was used for statistical analysis. Results: There was no statistically significant difference in SNA, SNB, ANB and SNPg between the centres at the age of 6 years. There were no children with a class III jaw relationship. The sagittal dimensions were close to the values of non-cleft control persons (Droschl standards). Conclusion: There was considerable similar sagittal growth of the facial skeleton in both centres which has not been affected by the different surgical protocols so far. A final evaluation should be delayed until the growth of the facial skeleton is complete. © 2007 European Association for Cranio-Maxillofacial Surgery

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INTRODUCTION

Midfacial growth was described in detail by *Enlow* (1990). Studies of adult patients with unoperated complete unilateral cleft lip, alveolus and palate (UCLAP) indicate that there is a normal potential for maxillary growth (*Shetye*, 2004). Many influences on growth disturbances of the midface are described. The way of breathing influences growth of the midface in non-cleft children as well as in children with orofacial clefts (*Linder-Aronson et al.*, 1993). Proper resting posture of the oral cavity i.e., the mandible, tongue and cheeks seems to ensure normal growth of the maxilla and the sagittal intermaxillary relationship in patients with UCLAP (*Koželj and Drevenšek*, 2003). The differing severity of clefting and normal biological variations of the craniofacial complex (*Molsted*, 1999; *Peltomäki et al.*, 2001) may determine the results of maxillary growth, as well as the lack of intrinsic growth potential (*Honda et al.*, 2002). Presurgical orthopaedics and extraoral strapping are controversial. Some authors see positive effects using presurgical orthopaedic treatment by reducing the width of the cleft and therefore making subsequent surgery easier (*Salzer*, 1997; *Yamada et al.*, 2003). *Molsted et al.* (1992), however, did not find

a better treatment outcome after using presurgical orthopaedics. Surgery with resulting scars seems to be recognized as the main reason for poor growth of the maxillary region (*Herfert*, 1958; *Mars and Houston*, 1990; *Li et al.*, 2006). There are many different protocols concerning the timing and technique of surgery. Some authors think that early proper repair of the perioral and perinasal muscles activates normal physiological skeletal growth (*Talmant*, 1993; *Joos*, 1995). The timing and technique of palatal closure are discussed controversially in the literature (*Schultes et al.*, 2000; *Gaggl et al.*, 2003). The main advantage of early palatal closure is that it avoids the development of compensatory speech in early life (*Nguyen and Sullivan*, 1993), the disadvantage is growth impairment of the facial skeleton (*Rohrich et al.*, 1996). In a two-stage protocol the soft and the hard palate are closed by two separate procedures at different times. Following this philosophy the closure of the hard palate should be performed as late as possible. As was mentioned before, the main reason for preferring this technique is to avoid growth disturbances and scarring of the hard palate (*Schmid et al.*, 1974; *Lilja et al.*, 2006), the main disadvantage being a compensatory articulation (*Wu et al.*, 1988). *Gaggl et al.* (2003), however, could not find any